

Template Week 5 – Operating Systems

Student number: 587890

Assignment 5.1: Unix-like

- a) Find out what the difference is between UNIX and unix-like operating systems?
Unix means that it is certified by the open group, unix-like operating systems is designed similar
But not certified
- b) Study the image above named UNIX timeline. Find out who Ken Thompson, Dennis Ritchie, Bill Joy, Richard Stallman, and Linus Torvalds are and what they have contributed to the development of UNIX or unix-like systems and to IT in general. **TIP!** English-language sources often contain more detailed information about these individuals.

Ken Thompson, Co-creator of UNIX (1969) at Bell Labs. Created the B programming language, which influenced C. Designed the original UNIX file system and concepts like pipes.

Dennis Ritchie. Co-creator of UNIX with Thompson. Invented the C programming language.

Bill Joy, Co-founder of Sun Microsystems. Key developer of BSD UNIX. Major contributor to TCP/IP integration in UNIX.

- c) What is the philosophy of the GNU movement?
(0) to run the program,
(1) to study and change the program in source code form,
(2) to redistribute exact copies
(3) to distribute modified versions.
- d) Does Ubuntu as a Linux operating system conform to the philosophy of the GNU movement?
Please explain your answer.
How Ubuntu conform to the GNU philosophy
Ubuntu is based on GNU/Linux.
Most of its core components are free software.
Where Ubuntu conflicts with GNU ideals
Ubuntu includes some proprietary drivers (NVIDIA, Wi-Fi firmware).
Sometimes includes closed-source software repositories.
Provides Snap packages, some of which are not open-source.
Ships with a non-free kernel firmware by default.
- e) Find out what is the Windows Subsystem for Linux?
It's some thing you can use to run linux in windows with out a vm
- f) Find out, which operating system family belongs to Android, iOS and ChromeOS?
It's build on the linux kernel

Assignment 5.2: Supercomputers and gameconsoles

- a) Research on this site what supercomputers are used for and write a short summary of it:

<https://www.computerhistory.org/timeline/search/?q=Supercomputer>

Supercomputers are extremely powerful computers designed to perform vast numbers of calculations at very high speeds. A

- b) IBM is a company that has already built a number of supercomputers. One of them is IBM's Roadrunner. The CPU developed for this supercomputer was further developed at a later stage as the CPU for the PlayStation 3 console. Find out what a **PlayStation 3 cluster** is and what it was used for? The it is a collection of ps3 used as a parallel computing system working on big calculations, I has been used for medical and scientific research

- c) You can build a supercomputer by putting a few computers together in a cluster. Here's what Oracle did with a collection of Raspberry Pi's, for example:

<https://blogs.oracle.com/developers/post/building-the-worlds-largest-raspberry-pi-cluster>

What specific operating system is running on this cluster?

Oracle Linux for arm

- d) Does Oracle's Raspberry Pi supercomputer appear in the list of the 500 fastest supercomputers in the world? Make a logical decision for this, without going through the entire list.

<https://www.top500.org/lists/top500/list/2023/06/>

Nope the top500 list ranks systems based on raw performance, measured in FLOPS. And you would need an obscene amount of Raspberry's to get to that performance level

- e) What CPU architecture is used for the PlayStation 5 and Xbox Series X? x86-64 for both
What operating systems run on these consoles?

Xbox Series X: Xbox System Software (based on a customized Windows kernel)

PlayStation 5: A customized version of Orbis OS, based on FreeBSD (Most likely)

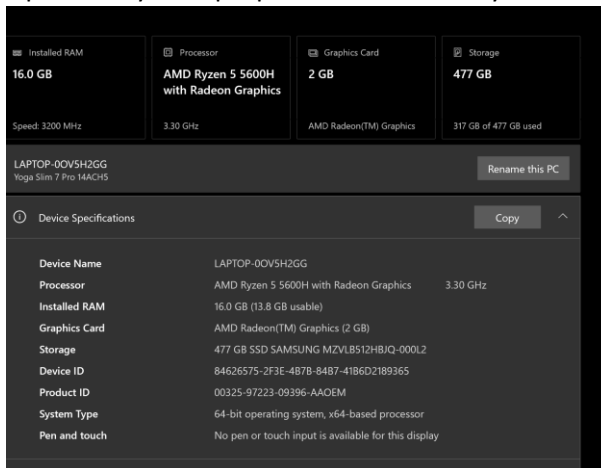
What conclusion can you draw from the answer to the previous question?

The game consoles run ,albeit modified, version's of normal desktop OSes

Assignment 5.3: Working with Windows

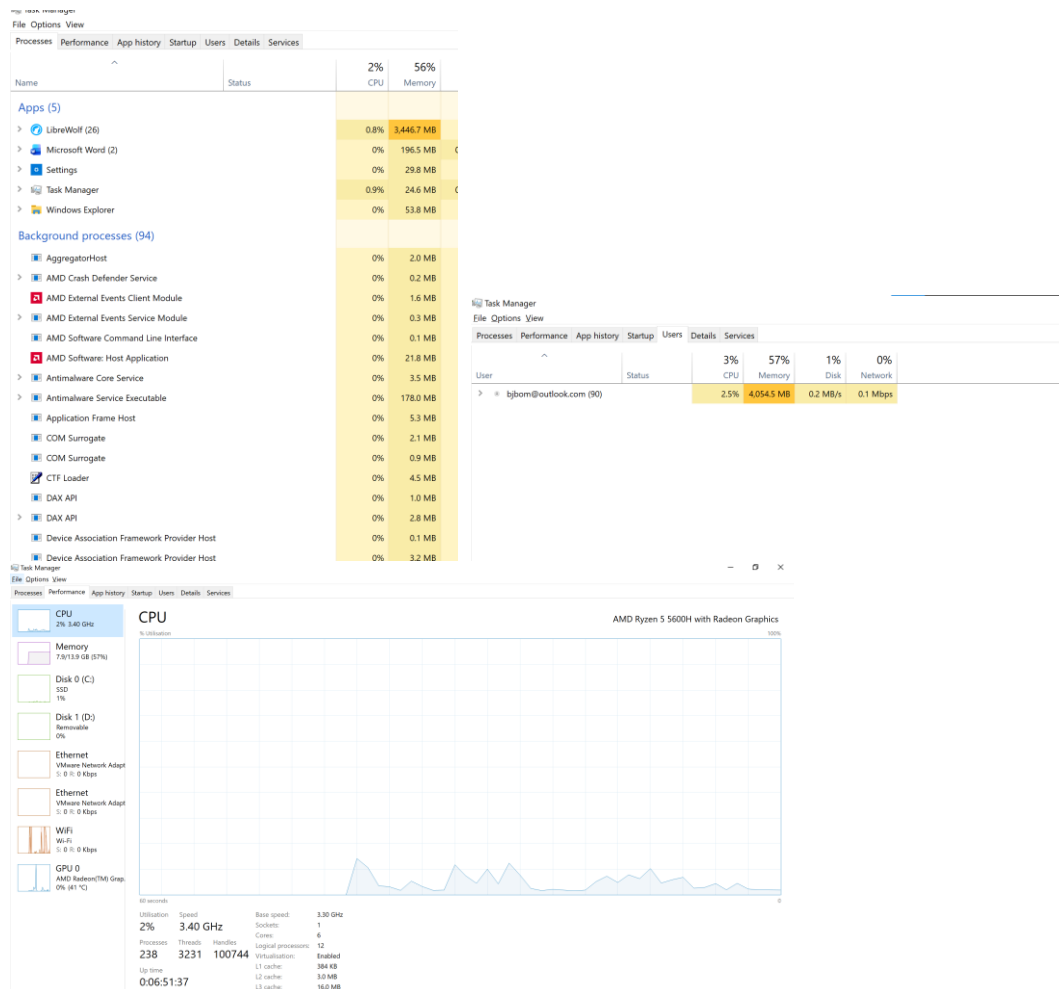
Take relevant screenshots of the assignments below

- Practice for about 10 minutes with the  keyboard shortcuts combinations, skip the general shortcuts in this exercise. Take a look at which screens are opened.
Start Menu, Connect screen Accessibility Center, the Action screen, the screen Lock, the Shortcuts menu, the System tray, the Emoji window the window Explorer, Run boxes, Game bar, Search Tab, Task viewer and Settings
- The file explorer can be opened with  + E, Which key combination could you also use? If you have it open CTRL + N
- Open the system properties with a  key combination, take a screenshot of the open screen.

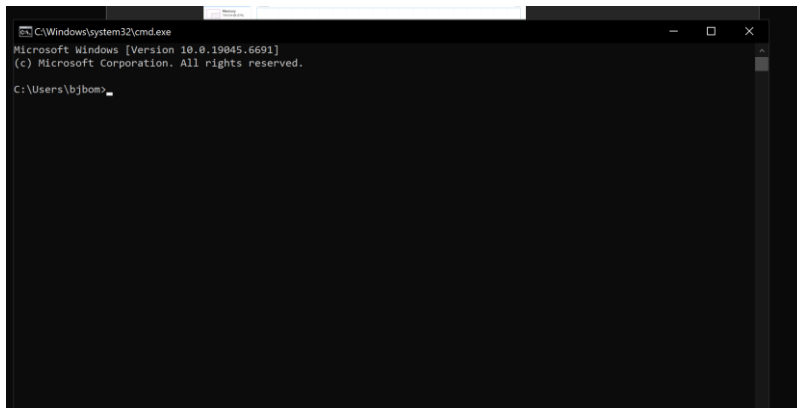


Paste this screenshot into this template.

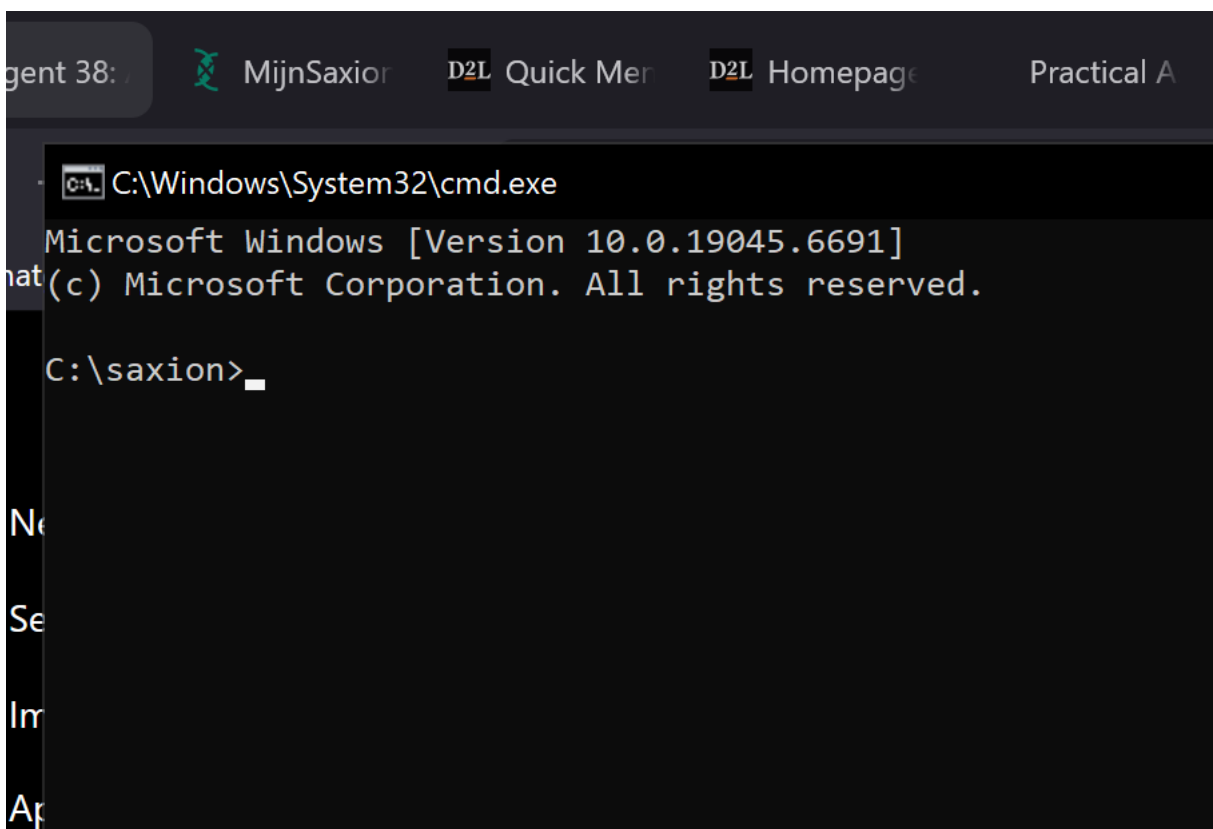
- Open task manager with a key combination. Take screenshots of the tabs: processes (shows active processes), performance, and users. Place these three screenshots in this template.



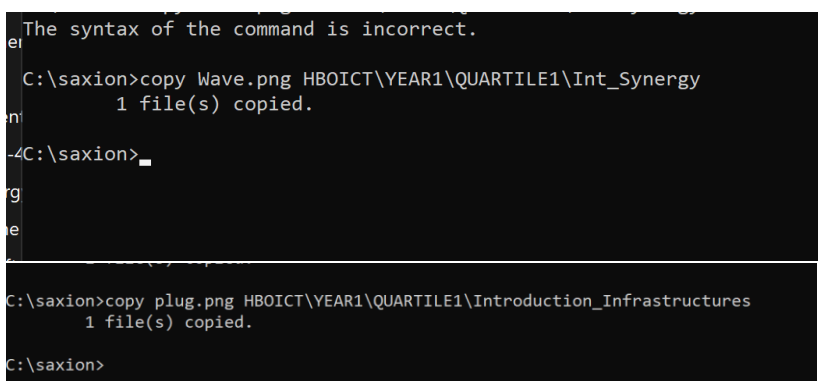
- e) If you're giving a PowerPoint presentation and you connect your laptop to a projector, Windows can use the projector as a second screen. For example, you may have Outlook open on your first screen that you don't show over the projector, while the PowerPoint presentation is displayed on the projector, or the second screen. Which key combination should you use for this? Windows + P
- f) If you leave the classroom for a while and you leave your laptop behind, it is wise to lock the screen. Your Apps will continue to run in the background. So, for example, if you're waiting for a download that takes a while, lock the screen and get a cup of coffee. Which key combination do you use for this? Windows + L
- g) Open the Run screen with a key combination. On this screen, type CMD and press <enter>. Take a screenshot of this result and paste it into this template.



Working in the File Explorer



Relevant screenshots **copy** command:



```
C:\saxion>copy plug.png HBOICT\YEAR1\QUARTILE1\Programming_Course
1 file(s) copied.
```

Relevant screenshots **tree** command:

```
C:\saxion>copy Tumble.png HBOICT\YEAR1\QUARTILE1\
C:\saxion>copy plug.png HBOICT\YEAR1\QUARTILE1\Programming_Course
1 file(s) copied.

C:\saxion>tree
Folder PATH listing for volume Windows-SSD
Volume serial number is 06BB-B2A9
C:.
|-- Debian 12.x 64-bit
|-- funny bird v2
|   |-- 564d486f-3563-7844-a451-a953ed375b38.vmem.lck
|   |-- Ubuntu 64-bit.vmdk.lck
|   |-- Ubuntu 64-bit.vmx.lck
|-- HBOICT
|   |-- YEAR1
|   |   |-- QUARTILE1
|   |   |   |-- Int Synergy
|   |   |   |-- Introduction Infrastructures
|   |   |   |-- Introduction to Programming
|   |   |-- QUARTILE2
|   |   |-- QUARTILE3
|   |   |-- QUARTILE4
|   |-- YEAR2
|   |   |-- QUARTILE1
|   |   |-- QUARTILE2
|   |   |-- QUARTILE3
|   |   |-- QUARTILE4
|   |-- YEAR3
|   |-- YEAR4
|-- iso- Bjarne
|-- vmware
|   |-- bin
|   |-- ico
|   |-- messages
|   |   |-- ja
|   |   |-- zh_CN
|   |-- OVFTool
|   |   |-- env
|   |   |   |-- en
|   |   |-- schemas
|   |   |   |-- DMTF
|   |   |   |-- vmware
|   |-- Resources
|   |-- tools-upgraders
|   |-- vkd
|   |-- x64
C:\saxion>echo bjarne
bjarne
```

Relevant screenshots in the file explorer of the folder c:\Saxion + created zip file.



HBOICT.rar

Terminating Processes

Relevant Screenshots Task Manager Window:

File Options View						
Processes Performance App history Startup Users Details Services						
Name	Status	6% CPU	85% Memory	0% Disk	0% Network	1% GPU
Apps (8)						
➤  Calculator (2)						
		0.1%	48.7 MB	0 MB/s	0 Mbps	0.1%

Install Software

Relevant screenshots that the following software is installed with winget:

- Firefox
- WinSCP
- Notepad++
- 7zip

```
C:\Users\bjbom>winget install -e --id Mozilla.Firefox
The Microsoft Store requires that you view the following agreements before using.
Terms of Transaction: https://aka.ms/microsoft-store-terms-of-transaction
The source requires the current machine's 2-letter geographic region to be sent to the backend service to function properly (ex. "US").
Do you agree to all the source agreements terms?
[V] Yes [N] No: yes
Found Mozilla Firefox (en-US) [Mozilla.Firefox] Version 146.0.1
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://download-installer.cdn.mozilla.net/pub/firefox/releases/146.0.1/win64/en-US/Firefox%20Setup%20146.0.1.exe
82.3 MB / 82.3 MB
Successfully verified installer hash
Starting package install...
Successfully installed
C:\Users\bjbom>
```

```
C:\Users\bjbom>winget install -e --id WinSCP
No package found matching input criteria.

C:\Users\bjbom>winget install -e --id 7zip.7zip
Found 7-Zip [7zip.7zip] Version 25.01
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://7-zip.org/a/7z2501-x64.exe
1.56 MB / 1.56 MB
Successfully verified installer hash
Starting package install...
The installer will request to run as administrator. Expect a prompt.
Successfully installed
C:\Users\bjbom>
```

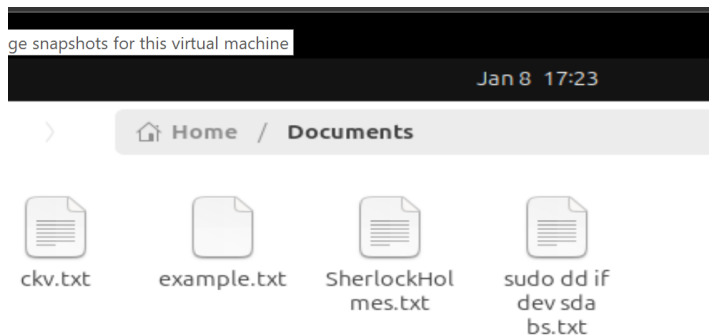
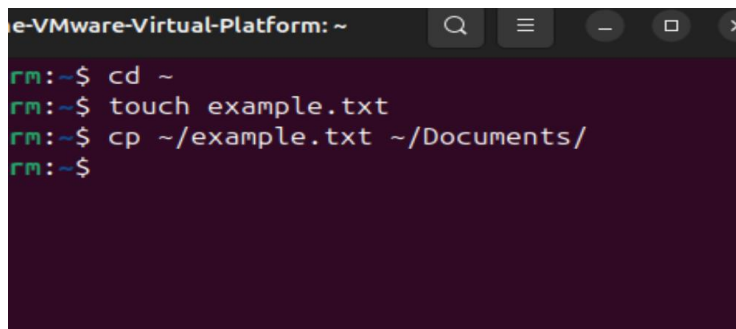
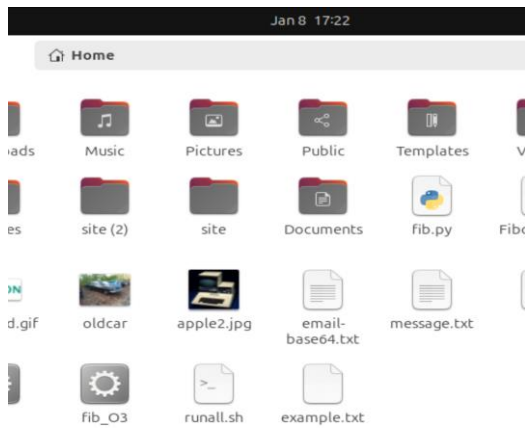
```
The installer will request to run as administrator. Expect a prompt.
Successfully installed

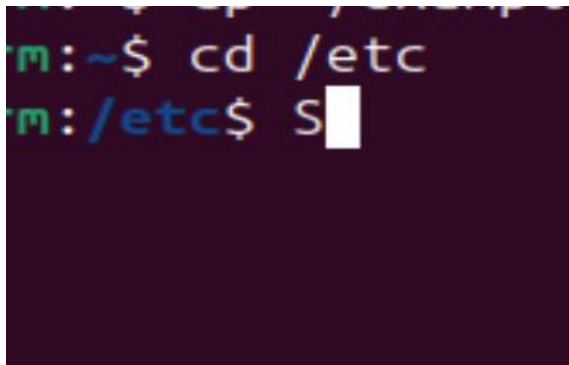
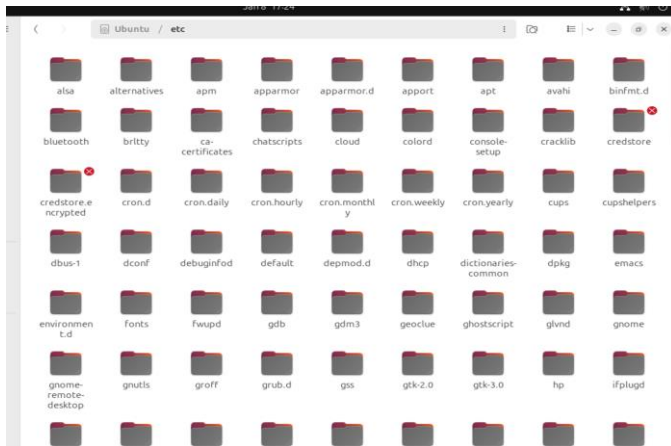
C:\Users\bjbom>winget install -e --id Notepad++.Notepad++
Found Notepad++ [Notepad++.Notepad++] Version 8.9
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://github.com/notepad-plus-plus/notepad-plus-plus/releases/download/v8.9/npp.8.9.Installer.x64.exe
6.54 MB / 6.54 MB
Successfully verified installer hash
Starting package install...
The installer will request to run as administrator. Expect a prompt.
Successfully installed
C:\Users\bjbom>
```

```
C:\Users\bjbom>winget install -e --id WinSCP.WinSCP
Found WinSCP [WinSCP.WinSCP] Version 6.5.5
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://sourceforge.net/projects/winscp/files/WinSCP/6.5.5/WinSCP-6.5.5-Setup.exe/download
11.6 MB / 11.6 MB
Successfully verified installer hash
Starting package install...
The installer will request to run as administrator. Expect a prompt.
Successfully installed
C:\Users\bjbom>
```

Assignment 5.4: Working with Linux

Relevant screenshots + motivation

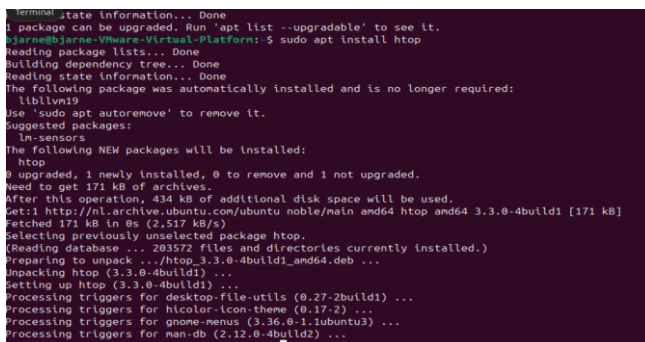
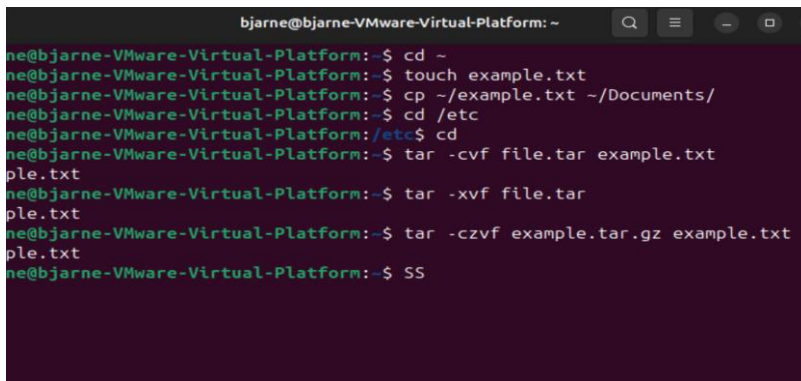


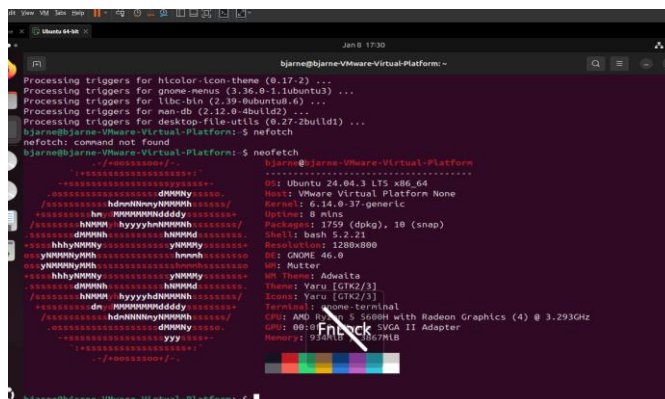


Linux has one single root directory (/), and all files and disks are mounted under it.

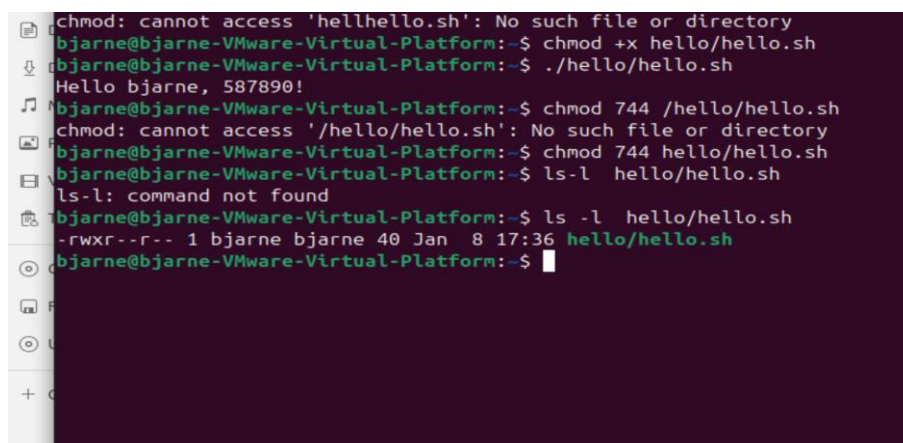
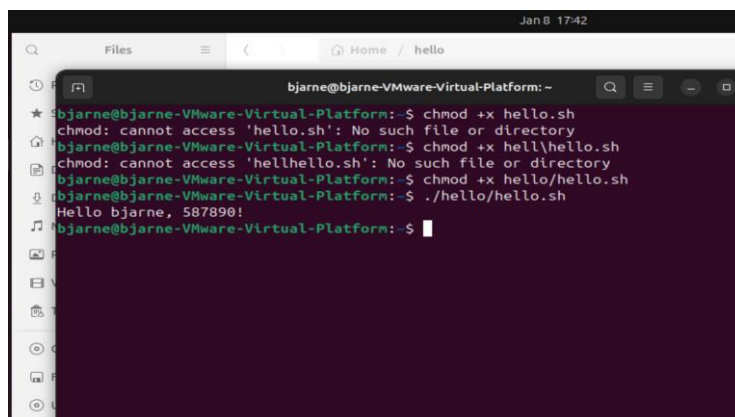
Windows uses multiple drive letters (C:, D:, E:).

The /etc directory contains system-wide configuration files, such as settings for users, services, networking, and installed software.





Relevant screenshots + motivation



Assignment 5.6: View the contents of files

Relevant screenshots + motivation

cat

Displays the entire contents of a file on the terminal at once.

wc (word count)

Counts lines, words, and characters in a file.

wc -l → lines

wc -w → words

wc -c → characters

less

Opens a file in a scrollable viewer (use ↑ ↓, PageUp/PageDown).

Press q to quit.

head

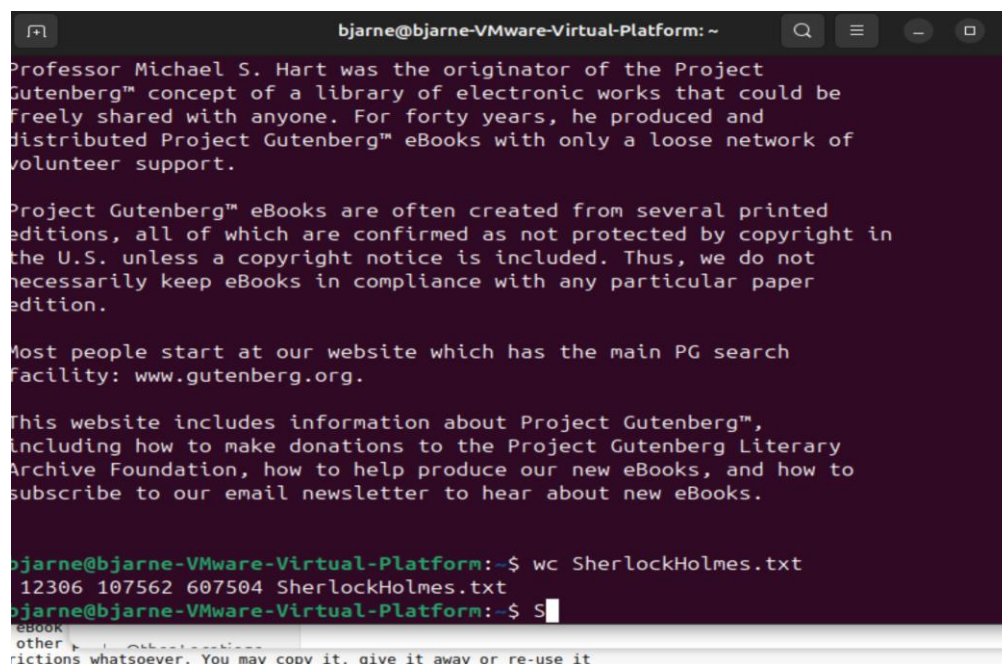
Shows the first part of a file (default: first 10 lines).

tail

Shows the last part of a file (default: last 10 lines).

grep

Searches for text inside a file and prints matching lines.



```
bjarne@bjarne-VMware-Virtual-Platform: ~  
Professor Michael S. Hart was the originator of the Project  
Gutenberg™ concept of a library of electronic works that could be  
freely shared with anyone. For forty years, he produced and  
distributed Project Gutenberg™ eBooks with only a loose network of  
volunteer support.  
  
Project Gutenberg™ eBooks are often created from several printed  
editions, all of which are confirmed as not protected by copyright in  
the U.S. unless a copyright notice is included. Thus, we do not  
necessarily keep eBooks in compliance with any particular paper  
edition.  
  
Most people start at our website which has the main PG search  
facility: www.gutenberg.org.  
  
This website includes information about Project Gutenberg™,  
including how to make donations to the Project Gutenberg Literary  
Archive Foundation, how to help produce our new eBooks, and how to  
subscribe to our email newsletter to hear about new eBooks.  
  
bjarne@bjarne-VMware-Virtual-Platform:~$ wc SherlockHolmes.txt  
12306 107562 607504 SherlockHolmes.txt  
bjarne@bjarne-VMware-Virtual-Platform:~$ S
```

```

bjarne@bjarne-VMware-Virtual-Platform:~$ wc SherlockHolmes.txt
12306 107562 607504 SherlockHolmes.txt
bjarne@bjarne-VMware-Virtual-Platform:~$ grep -n "kingdom" SherlockHolmes.txt
90:"I tell you that I would give one of the provinces of my kingdom to
124:And that was how a great scandal threatened to affect the kingdom of
bjarne@bjarne-VMware-Virtual-Platform:~$ S

```

```

eBook
other
ictions whatsoever. You may copy it, give it away or re-use it
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or online at www.gutenberg.org. If you are not located in the

490:"I tell you that I would give one of the provinces of my kingdom to
1124:And that was how a great scandal threatened to affect the kingdom of
bjarne@bjarne-VMware-Virtual-Platform:~$ grep -n -C 10 "kingdom" SherlockHolmes.
cxt
480-"Then I shall drop you a line to let you know how we progress."
481-
482-"Pray do so. I shall be all anxiety."
483-
484-"Then, as to money?"
485-
486-"You have _carte blanche_."
487-
488-"Absolutely?"
489-
490:"I tell you that I would give one of the provinces of my kingdom to
491-have that photograph."
492-
493-"And for present expenses?"
494-
495-The King took a heavy chamois leather bag from under his cloak and laid
496-it on the table

```

Assignment 5.7: Digital forensics

Relevant screenshots + motivation

```

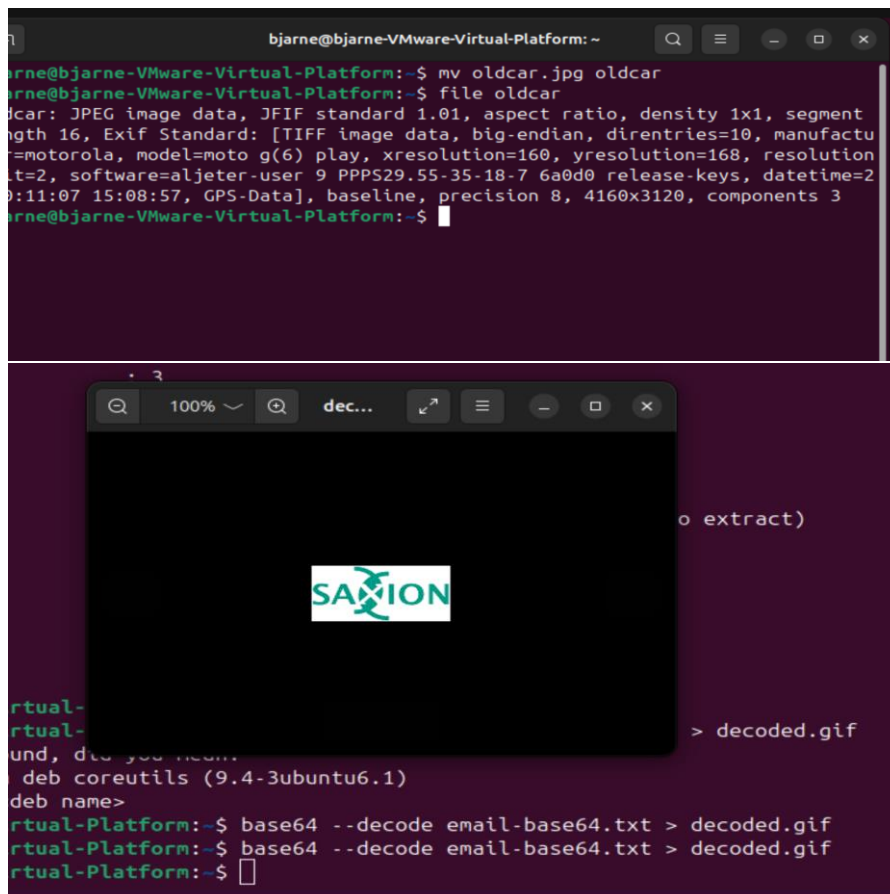
: -rw-r--r--
: JPEG
: jpg
: image/jpeg
: 1.01
: Big-endian (Motorola, MM)
: motorola
: moto g(6) play
: 72
: 72
: inches
: aljeter-user 9 PPPS29.55-35-18-7
: 2020:11:07 15:08:57

```

```

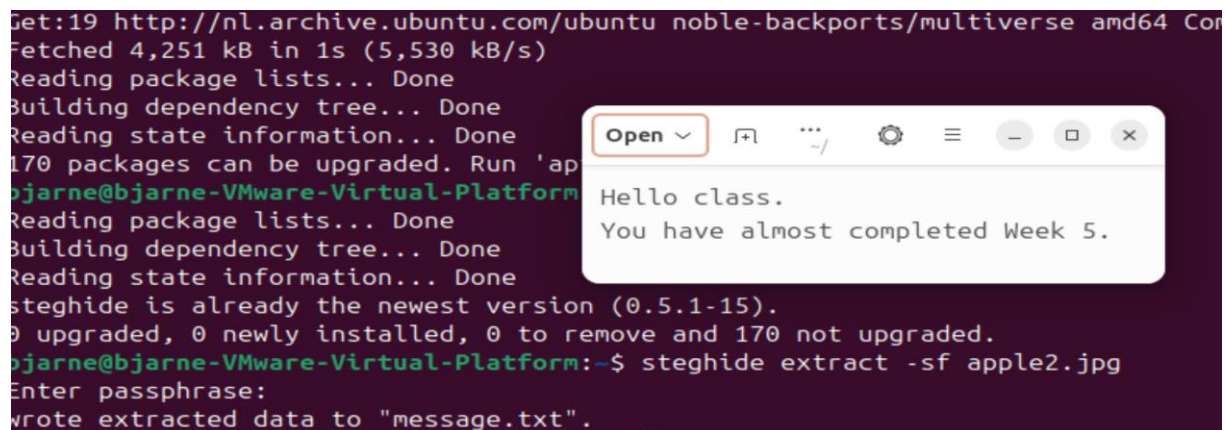
: 2020:11:07 14:08:57Z
: 53 deg 11' 39.68" N
: 6 deg 32' 12.90" E
: 3.5 mm
: 53 deg 11' 39.68" N, 6 deg 32' 12.90" E
: 7.7
Platform: 6 AC

```

Assignment 5.8: Steganography

Relevant screenshots + motivation



Assignment 5.9: Capture disk images

Make relevant screenshots + motivation:

- Proof that the Debian 13 server stored a back-up image of the Ubuntu 24.04 Desktop VM.
- Proof that you can restore the back-up image into an empty VM.

```

ermitted by applicable law.
bjarne@bjarne:~$ sudo apt update
sudo: password for bjarne:
Hit:1 http://deb.debian.org/debian bookworm InRelease
Hit:2 http://security.debian.org/debian-security bookworm-security InRelease
Hit:3 http://deb.debian.org/debian bookworm-updates InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
bjarne@bjarne:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:9.2p1-2+deb12u7).
openssh-server set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
bjarne@bjarne:~$ sudo systemctl enable --now ssh
Synchronizing state of ssh.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable ssh

```

```

user@192.168.139.133's password:
| Files | sion denied, please try again.
user@192.168.139.133's password:
Permission denied, please try again.
user@192.168.139.133's password:
user@192.168.139.133: Permission denied (publickey,password).
bjarne@bjarne-VMware-Virtual-Platform:~$ ssh bjarne@192.168.139.133
bjarne@192.168.139.133's password:
Permission denied, please try again.
bjarne@192.168.139.133's password:
Permission denied, please try again.
bjarne@192.168.139.133's password:
Linux bjarne 6.1.0-41-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.158-1 (2025-11-09)
x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Jan  7 11:28:20 2026
bjarne@bjarne:~$

```

```

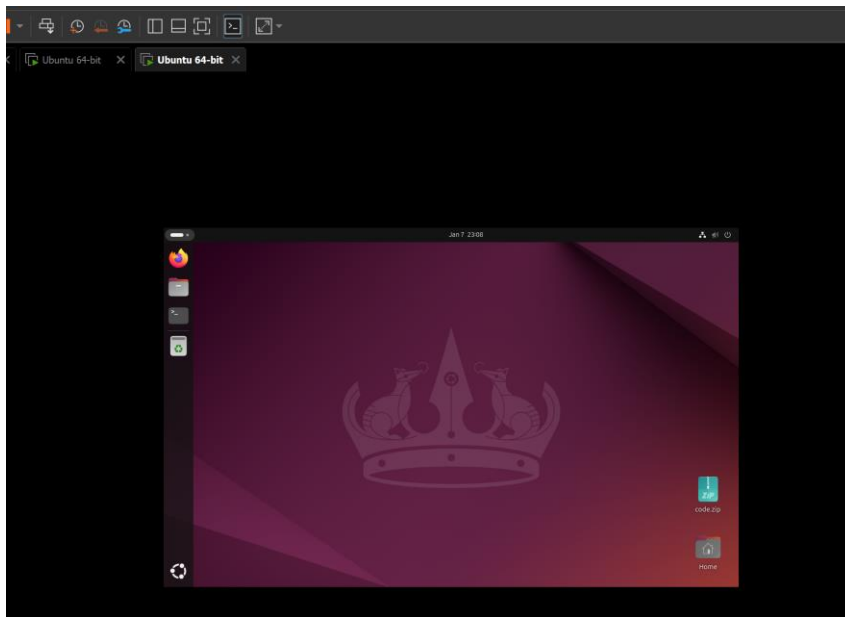
bjarne@bjarne-VMware-Virtual-Platform:~$ sudo dd if=/dev/sda bs=4M status=progress | gzip | ssh bjarne@192.168.139 "cat > /srv/images/ubuntu2404_vm.img.gz"
do: password for bjarne:
: connect to host 192.168.0.139 port 22: Connection refused
bjarne@bjarne-VMware-Virtual-Platform:~$ sudo dd if=/dev/sda bs=4M status=progress | gzip | ssh bjarne@192.168.139.133 "cat > /srv/images/ubuntu2404_vm.img.gz"
bjarne@192.168.139.133's password:
212672 bytes (285 MB, 272 MiB) copied, 13 s, 21.7 MB/s

```

```

Jan 7 22:00
ubuntu@ubuntu:~$ [200-ssh bjarne@192.163.139.133 "cat /srv/images/ubuntu2404_vm.img.gz" | gzip -d | sudo dd of=/dev/sda
a bs=4M status=progress
ssh: command not found
gzip: stdin: unexpected end of file
0+0 records in
0+0 records out
0 bytes copied, 0.214831 s, 0.0 kB/s
ubuntu@ubuntu:~$ ssh bjarne@192.163.139.133 "cat /srv/images/ubuntu2404_vm.img.gz" | gzip -d | sudo dd of=/dev/sda bs=4M
status=progress
ssh: connect to host 192.163.139.133 port 22: Connection refused
gzip: stdin: unexpected end of file
0+0 records in
0+0 records out
0 bytes copied, 21.0388 s, 0.0 kB/s
ubuntu@ubuntu:~$ ssh bjarne@192.168.139.133 "cat /srv/images/ubuntu2404_vm.img.gz" | gzip -d | sudo dd of=/dev/sda bs=4M
status=progress
The authenticity of host '192.168.139.133 (192.168.139.133)' can't be established.
ED25519 key fingerprint is SHA256:8Nucr4Rq7738ZKYGc75o1K1JkGwP31ax/B998F8xQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.139.133' (ED25519) to the list of known hosts.
bjarne@192.168.139.133's password:
54735232 bytes (545 MB, 529 MiB) copied, 15 s, 36.3 MB/s

```



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